

Sustainable Software Engineering Software Engineering for Artificial Intelligence

GROUP PROJECT



Sustainable Software Engineering - The Project of the Course

Starting Point

- Based on a **previous project** (from the Software Engineering course).
- The project must be a **web application** that **used Git** for version control.
- Group Project 2/3 people

Goal

Apply the concepts and tools learned in this course to make your project more sustainable — technically, socially, and environmentally — using the tools introduced during the lectures. If your project includes an AI component, evaluate its fairness metrics.

Future Integration

This project can be extended for **SE4AI (Software Engineering for AI)** next semester, where you'll **build a ML pipeline inside the project or re-engineer the one you already have**, focusing on **MLOps**, based on the project you choose in this course.

Sustainable Software Engineering: Deliverables

1. **Technical Report (main artifact) - Before the discussion**

Description of the original project and sustainability goals.

Tools applied and methodology used.

Results of energy, performance, and social analyses.

Discussion of improvements and limitations.

Optional: Fairness metrics (if AI included).

2. **Practical Evidence (Inside the report)**

Screenshots, logs, or outputs from tools.

Graphs or tables of measurements (before/after).

Code snippets showing refactoring or improvements.

3. **Git Repository (Link)**

Full commit history.

4. **Reflection / Sustainability Statement (Inside the report - 1 page)**

What did you learned about sustainable software engineering?

What trade-offs did you face (e.g., performance vs. maintainability)?

How does this project connect to SE4AI or long-term software quality?

5. **Presentation - Discussion (10-15 min)**

Focused on process, findings, and key insights.

Optional live demo of tool usage or performance measurement.

Sustainable Software Engineering Software Engineering for Artificial Intelligence

GROUP PROJECT

